Methods for Portfolio Optimisation - Group 31

Sara Scully, 2194423; Davide Giardini, 2202652

Contents

1 Abstract **2**

2 Introduction **2**

3 Markowitz Classic portfolio problem **2**
3.1 Proof for gradient of the Markowitz . . . 3

4 Frank-Wolfe Methods - Theory **4**
4.1 Search Direction Analysis . . . 5
4.1.1 Proof Of Direction $\nabla f(x_k)^\top(\hat{x}_k - x_k) > 0$ . . . 5
4.2 Frank-Wolfe Convergence Theorem . . . 6
4.2.1 Line search for Portfolio Optimisation . . . 7
4.2.2 Diminishing Step Size . . . 8
4.2.3 Stopping Criterion via Duality . . . 8
4.3 Away-Step Frank-Wolfe (ASFW) . . . 9
4.4 Pairwise Frank-Wolfe (PWFW) . . . 10
4.5 Convergence Properties . . . 11

5 Description of Datasets **11**
5.1 Return Characteristics . . . 12
5.2 Correlation . . . 12

6 Results **13**
6.1 Convergence Rate . . . 13
6.2 Objective Value . . . 14
6.3 Duality Gap . . . 15
6.4 Final Active Set Size . . . 15
6.5 Execution Time . . . 16
6.6 Diminishing Step Size vs Exact Line Search . . . 16
6.7 Comparison with Projected Gradient method . . . 17
6.8 Restricting to last 595 Time Periods . . . 18

7 Further Analysis **18**
7.1 Rolling-Window Portfolio Optimisation . . . 18
7.1.1 Risk Parameter Analysis . . . 19
7.1.2 Rolling-Window Optimisation Results . . . 20

8 Conclusion **21**

9 LLM Usage Disclosure Statement **21**

1 Abstract

Portfolio optimisation is a fundamental problem in finance, where the objective is to allocate capital across assets while balancing expected return and portfolio risk. This project investigates the use of Frank-Wolfe (FW), Away-Step Frank-Wolfe (ASFW), and Pairwise Frank-Wolfe (PWFW) methods for solving the Markowitz portfolio optimisation problem. The algorithms are evaluated using weekly return data from the Dow Jones, FTSE 100, and S&P 500 indices, containing 28, 83, and 442 assets respectively. The results show that all three methods achieve very similar objective values and sparse portfolios. However, ASFW and PWFW provide much faster convergence for larger datasets. Standard FW requires 4,172 iterations for FTSE 100 and 3829 for S&P 500, compared with only 11-16 iterations for ASFW and PWFW. The project also analyses exact line search versus diminishing step sizes, showing that their performance depends on the dataset. A comparison with Projected Gradient Descent (PGD) shows that PGD can also obtain accurate solutions, although the simplex projection is more computationally expensive than the FW linear minimisation oracle. Finally, a rolling-window analysis examines different risk-aversion parameters, with PWFW showing particularly strong performance on the S&P 500. Overall, the results indicate that active-set Frank-Wolfe methods are particularly advantageous for larger portfolio optimisation problems.

2 Introduction

Portfolio optimisation is a central problem in finance, aiming to determine how capital should be allocated across assets to achieve a suitable balance between risk and return. The classical Markowitz portfolio optimisation model formulates this as a convex quadratic problem, where the investor's risk preference is controlled by a risk-aversion parameter.

This project investigates the use of Frank-Wolfe (FW), Away-Step Frank-Wolfe (ASFW), and Pairwise Frank-Wolfe (PWFW) to solve this problem under no-short-selling and budget constraints. Frank-Wolfe methods are particularly suitable for portfolio optimisation because they avoid expensive projections onto the simplex by using a linear minimisation oracle. However, standard FW can converge slowly and exhibit zig-zagging near boundary solutions. This motivates the use of its active-set variants.

The methods are evaluated on Dow Jones, FTSE 100, and S&P 500 datasets, with further analysis of step-size strategies, comparison with Projected Gradient Descent (PGD), and a rolling-window framework examining different levels of risk aversion. The aim is to assess the convergence, computational efficiency, and practical performance of the different optimisation methods.

3 Markowitz Classic portfolio problem

Let us consider an investor with a budget of B (we normalise this to $B = 1$) who can invest in n different assets. Over a given period let:

- x_i denote the proportion of wealth invested in asset i during the period, $i = 1, 2, \ldots, n$
- r_i be a random variable denoting the return of asset i over a period of the given length
- $\bar{r}_i = \mathbb{E}[r_i]$ be the expected return over the period for asset i
- $\Sigma_{ij} = \mathrm{Cov}(r_i, r_j)$ be the covariance between the return of assets i and j

The portfolio is fully described by the weight vector $\mathbf{x} \in \mathbb{R}^n$, which must satisfy the budget constraint that all capital must be allocated, $\sum_{i=1}^{n} x_i = 1$ and the non-negativity constraint that no short selling is allowed, $x_i \geq 0$ for all i.

The return on the portfolio is given by a random variable

$$r_p = \sum_{i=1}^{n} x_i r_i$$

The expected return of the portfolio is given by:

$$\mathbb{E}[r_p] = \sum_{i=1}^{n} x_i \bar{r}_i = \bar{\mathbf{r}}^\top \mathbf{x}$$

where $\bar{\mathbf{r}} = \{\bar{r}_1, \bar{r}_2, \bar{r}_3, \dots \bar{r}_n\}$

and the variance (risk) of the portfolio is:

$$\text{Var}(r_p) = \sum_{i=1}^{n} \sum_{j=1}^{n} x_i x_j \Sigma_{ij} = \mathbf{x}^\top \Sigma \mathbf{x}$$

Assume the investor seeks to maximise expected return while minimising risk. Markowitz formulated this as a convex quadratic programming problem:

$$\max_{\mathbf{x} \in \mathbb{R}^n} \quad \bar{\mathbf{r}}^\top \mathbf{x} - \gamma \mathbf{x}^\top \Sigma \mathbf{x}$$

Which is equivalent to,

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^n} \quad & \gamma \mathbf{x}^\top \Sigma \mathbf{x} - \bar{\mathbf{r}}^\top \mathbf{x} \\ \text{s.t.} \quad & e^\top \mathbf{x} = 1 \\ & \mathbf{x} \geq 0 \end{aligned}$$

where $\gamma > 0$ is the risk aversion parameter for the investor. This parameter controls the trade-off between risk and return. A large γ penalises risk more heavily, producing less risky portfolios, suitable for a risk-averse investor, while a small γ places greater emphasis on expected return, resulting in more risky portfolios.

When $\gamma = 0$, the problem reduces to maximising expected return, which results in a portfolio concentrated in the single best-performing asset based on historical data if the maximiser is unique. As $\gamma \to \infty$, the portfolio converges to the global minimum variance portfolio.

3.1 Proof for gradient of the Markowitz

We start with the mean-variance objective function:

$$f(x) = \gamma \mathbf{x}^\top \Sigma \mathbf{x} - \bar{\mathbf{r}}^\top \mathbf{x}$$

We compute its gradient $\nabla f(x)$ to determine the search direction and step size required for the Frank-Wolfe algorithm. The gradient is a vector of dimension n.

Let $L(x) = \bar{r}^\top x = \sum_{i=1}^{n} \bar{r}_i x_i$. Taking the partial derivative with respect to x_k:

$$\frac{\partial L(x)}{\partial x_k} = \frac{\partial}{\partial x_k} \sum_{i=1}^{n} \bar{r}_i x_i = \bar{r}_k \implies \nabla L(x) = \bar{\mathbf{r}} = \{\bar{r}_1, \bar{r}_2, \bar{r}_3, \dots \bar{r}_n\}$$

Let $Q(x) = x^\top \Sigma x = \sum_{i=1}^{n} \sum_{j=1}^{n} \Sigma_{ij} x_i x_j$. Differentiating with respect to x_k:

$$\frac{\partial Q}{\partial x_k} = \frac{\partial}{\partial x_k} \left(\sum_{i=1}^{n} \sum_{j=1}^{n} \Sigma_{ij} x_i x_j \right)$$

When $i = k$ and $j \neq k$ the derivative with respect to x_k is $\Sigma_{kj} x_j$. When $j = k$ and $i \neq k$ the derivative with respect to x_k is $\Sigma_{ik} x_i$. When $j = k$ and $i = k$ the term is $\Sigma_{kk} x_k^2$, the derivative is $2\Sigma_{kk} x_k$. Finally,

when $i, j \neq k$, the term $\Sigma_{ij} x_i x_j$ does not depend on x_k, and therefore its partial derivative with respect to x_k is zero.

Combining these cases:

$$\frac{\partial Q}{\partial x_k} = 2\Sigma_{kk} x_k + \sum_{j=1,;j\neq k}^{n} \Sigma_{kj} x_j + \sum_{i=1,;i\neq k}^{n} \Sigma_{ik} x_i.$$

Since the covariance matrix Σ is symmetric ($\Sigma_{ik} = \Sigma_{ki}$):

$$\frac{\partial Q}{\partial x_k} = 2\Sigma_{kk} x_k + 2 \sum_{j=1,;j\neq k}^{n} \Sigma_{kj} x_j.$$

The gradient is a vector of dimension n. Combining the linear and quadratic components gives:

$$\nabla f(x) = 2\gamma \Sigma \mathbf{x} - \bar{\mathbf{r}}$$

4 Frank-Wolfe Methods - Theory

The Frank-Wolfe (FW) algorithm, also known as the Conditional Gradient method, is a first-order optimisation algorithm for solving constrained convex optimisation problems of the form

$$\min_{\mathbf{x}\in\mathcal{C}} f(\mathbf{x}),$$

where the feasible set $\mathcal{C} \subseteq \mathbb{R}^n$ is a compact convex set and $f : \mathbb{R}^n \to \mathbb{R}$ is a convex, differentiable function with an L-Lipschitz continuous gradient for some $L > 0$ where L is the Lipschitz constant of the gradient of f. That is of the form,

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \quad \forall x, y \in \mathbb{R}^n.$$

An important characteristic of the FW algorithm is that it is projection-free. This means that unlike projected gradient methods, requiring the computation of a projection onto the feasible set at every iteration, FW only requires the solution of a linear minimisation problem over $\mathcal{C}$. This results in a lower computational cost per iteration.

Since $\mathcal{C}$ is compact and f is continuous, the Weierstrass Theorem guarantees that the optimisation problem has at least one global minimiser. The algorithm therefore starts from a feasible point, denoted by $x_k \in \mathcal{C}$.

We wish to solve,

$$\arg\min_{x\in\mathcal{C}} \nabla f(x_k)^\top (x - x_k).$$

$$= \arg\min_{x\in\mathcal{C}} \nabla f(x_k)^\top x.$$

$$= \arg\min_{x\in\mathcal{C}} \left[f(x_k) + \nabla f(x_k)^\top (x - x_k) \right].$$

Since $\mathcal{C}$ is compact and the objective of the linear minimisation problem is continuous, there exists at least one solution $\hat{x}_k \in \mathcal{C}$

4.1 Search Direction Analysis

By definition of $\hat{x}_k$

$$\nabla f(x_k)^\top (\hat{x}_k - x_k) \leq \nabla f(x_k)^\top (x - x_k), \qquad \forall x \in \mathcal{C}.$$

We now distinguish between two possible cases.

First we consider the case when,

$$\nabla f(x_k)^\top (\hat{x}_k - x_k) = 0.$$
$$\nabla f(x_k)^\top (\hat{x}_k - x_k) \leq \nabla f(x_k)^\top (x - x_k), \qquad \forall x \in \mathcal{C},$$

Therefore,

$$\nabla f(x_k)^\top (x - x_k) \geq 0, \qquad \forall x \in \mathcal{C}.$$

This is the first-order optimality condition for the constrained convex optimisation problem. Therefore, $\hat{x_k}$ is a global minimiser of f over $\mathcal{C}$, and the algorithm can terminate.

We consider the second case, when

$$\nabla f(x_k)^\top (\hat{x}_k - x_k) < 0.$$

In this case, we define

$$d_k = \hat{x}_k - x_k.$$

Since

$$\nabla f(x_k)^\top d_k < 0,$$

d_k is a descent direction at x_k. Therefore, we can move from x_k in the direction d_k to obtain a new iterate

$$x_{k+1} = x_k + \alpha_k d_k,$$

where $\alpha_k \in (0, 1]$ is determined by a suitable line-search procedure.

4.1.1 Proof Of Direction $\nabla f(x_k)^\top (\hat{x}_k - x_k) > 0$

Note that the case $\nabla f(x_k)^\top (\hat{x}_k - x_k) > 0$ is impossible.

By definition, the Frank-Wolfe subproblem defines $\hat{x}_k$ as a minimiser of the linearised function over the feasible set $\mathcal{C}$

$$\hat{x}_k = \arg\min_{x \in \mathcal{C}} \nabla f(x_k)^\top x$$

Since we have that $\hat{x}_k$ minimises $\nabla f(x_k)^\top x$ over all $x \in \mathcal{C}$, its value cannot exceed the value evaluated at x_k:

$$\nabla f(x_k)^\top \hat{x}_k \leq \nabla f(x_k)^\top x_k$$

When we subtract

$$\nabla f(x_k)^\top x_k$$

from both sides we get the resulting inequality

$$\nabla f(x_k)^\top (\hat{x}_k - x_k) \leq 0$$

Therefore since $\nabla f(x_k)^\top (\hat{x}_k - x_k)$ is always not positive the case $\nabla f(x_k)^\top (\hat{x}_k - x_k) > 0$ cannot occur.

Algorithm 1 Frank-Wolfe Method [Jaggi, 2013]

Choose a point $x_1 \in C$
for $k = 1, 2, \ldots$ **do**
  Set $\hat{x}_k = \text{Argmin}_{x \in C} \nabla f(x_k)^\top (x - x_k)$
  if $\hat{x}_k$ satisfies some specific condition **then**
    STOP
  end if
  Set $x_{k+1} = x_k + \alpha_k(\hat{x}_k - x_k)$, with $\alpha_k \in (0, 1]$ suitably chosen
end for

4.2 Frank-Wolfe Convergence Theorem

Let f be a convex function with Lipschitz continuous gradient having constant $L > 0$. The Frank-Wolfe method with diminishing stepsize $\alpha_k = \frac{2}{k+2}$ satisfies the convergence bound,

$$f(x_{k+1}) - f(x^*) \leq \frac{2LD^2}{k+1}, \quad \forall k \geq 1,$$

where

$$D = \max_{x,y \in C} |x - y|$$

denotes the diameter of the feasible set C and x^* is the optimal solution for x. Thus, the Frank-Wolfe method achieves a sublinear convergence rate of $\mathcal{O}(1/k)$ in terms of the objective function error.

$$\left\{ \mathbf{x} \in \mathbb{R}^n \mid \mathbf{e}^\top \mathbf{x} = 1, \quad \mathbf{x} \geq 0 \right\}$$

is the standard simplex.

The feasible set $\mathcal{C}$ is a compact, convex polytope, and the objective $f(\mathbf{x})$ is a convex quadratic function, provided that Σ is positive semidefinite.

$$\arg\min_i [\nabla f(x_k)] i.$$

The solution to the linear subproblem is then the corresponding vertex of the simplex,

$$\hat{x}_k = e_{i_k},$$

where e_{i_k} denotes the i_k-th standard basis vector. Hence, the Frank-Wolfe algorithm for the unit simplex can be written as follows.

Algorithm 2 Frank-Wolfe Method for Unit Simplex

Choose an initial point $x_1 = e_i$ for some $i \in \{1, \ldots, n\}$
for $k = 1, 2, \ldots$ **do**
  Compute $i_k = \arg\min_i [\nabla f(x_k)]_i$
  Set $\hat{x}_k = e_{i_k}$
  if $\hat{x}_k$ satisfies the stopping criterion **then**
    STOP
  end if
  Set step size $\alpha_k = \frac{2}{k+2}$
  Update $x_{k+1} = x_k + \alpha_k(\hat{x}_k - x_k)$
end for

At each iteration, the linear minimisation step

$$i_k = \arg\min_i [\nabla f(x_k)]_i$$

acts as the linear minimisation oracle (LMO). Since the feasible region is the standard simplex, the minimum of a linear function over $\mathcal{C}$ is attained at one of its vertices. Therefore, the LMO returns

$$\hat{x}_k = e_{i_k},$$

where i_k corresponds to the smallest component of the gradient. This in turn allows the FW method to avoid a costly, explicit projection step while maintaining feasibility throughout the iterations.

4.2.1 Line search for Portfolio Optimisation

The portfolio objective function $f(x)$ is defined as:

$$f(x) = \gamma_{\text{risk}} \mathbf{x}^T \Sigma \mathbf{x} - \bar{\mathbf{r}}^T \mathbf{x}$$

where $\gamma_{\text{risk}} > 0$ is the risk aversion parameter, $\Sigma \in \mathbb{R}^{n \times n}$ is the symmetric covariance matrix, and $\bar{\mathbf{r}} \in \mathbb{R}^n$ is the expected returns vector. The first term represents the portfolio risk, weighted by the investor's risk-aversion parameter, while the second term represents the negative expected return. Therefore, minimising $f(x)$ balances the trade-off between portfolio risk and expected return.

The objective function is differentiable and its gradient is given by

$$\nabla f(x) = 2\gamma_{\text{risk}} \Sigma \mathbf{x} - \bar{\mathbf{r}}$$

Evaluating $f(x + \gamma d)$ along direction vector d with step size $\gamma \in [0, 1]$ we get:

$$f(x + \gamma d) = \gamma_{\text{risk}} (x + \gamma d)^T \Sigma (x + \gamma d) - \bar{\mathbf{r}}^T (x + \gamma d)$$

Expanding the quadratic term first:

$$\begin{aligned} (x + \gamma d)^T \Sigma (x + \gamma d) &= (x^T + \gamma d^T)(\Sigma x + \gamma \Sigma d) \\ &= x^T \Sigma x + \gamma x^T \Sigma d + \gamma d^T \Sigma x + \gamma^2 d^T \Sigma d \end{aligned}$$

Since Σ is symmetric, $d^T \Sigma x = (x^T \Sigma d)^T = x^T \Sigma d$. Combining the cross terms gives:

$$(x + \gamma d)^T \Sigma (x + \gamma d) = x^T \Sigma x + 2\gamma x^T \Sigma d + \gamma^2 d^T \Sigma d$$

Next, expanding the linear return term:

$$-\bar{\mathbf{r}}^T (x + \gamma d) = -\bar{\mathbf{r}}^T x - \gamma \bar{\mathbf{r}}^T d$$

Substituting both expanded terms back into $f(x + \gamma d)$:

$$\begin{aligned} f(x + \gamma d) &= \gamma_{\text{risk}} \left(x^T \Sigma x + 2\gamma x^T \Sigma d + \gamma^2 d^T \Sigma d \right) - \bar{\mathbf{r}}^T (x + \gamma d) \\ &= \gamma_{\text{risk}} x^T \Sigma x - \bar{\mathbf{r}}^T x + \gamma (2\gamma_{\text{risk}} x^T \Sigma d - \bar{\mathbf{r}}^T d) + \gamma^2 \gamma_{\text{risk}} d^T \Sigma d \end{aligned}$$

Next, grouping the expression by the powers of γ. For the constant term we get,

$$\gamma_{\text{risk}} \mathbf{x}^T \Sigma \mathbf{x} - \bar{\mathbf{r}}^T \mathbf{x} = f(x)$$

For the linear term, we result in,

$$\gamma \left(2\gamma_{\text{risk}} x^T \Sigma d - \bar{\mathbf{r}}^T d \right) = \gamma \left(2\gamma_{\text{risk}} \Sigma x - \bar{\mathbf{r}} \right)^T d = \gamma \langle \nabla f(x), d \rangle$$

Finally, for the quadratic term we establish that,

$$\gamma^2 \gamma_{\text{risk}} d^T \Sigma d$$

Collecting all powers yields the scalar quadratic formulation:

$$f(x + \gamma d) = f(x) + \gamma \langle \nabla f(x), d \rangle + \gamma^2 \gamma_{\text{risk}} d^T \Sigma d$$

Setting the derivative of $f(x + \gamma d)$ with respect to γ to zero:

$$\frac{\mathrm{d}}{\mathrm{d}\gamma} f(x + \gamma d) = \langle \nabla f(x), d \rangle + 2\gamma \gamma_{\text{risk}} d^T \Sigma d = 0$$

Solving for the unconstrained optimal step size γ^*:

$$\gamma^* = \frac{-\langle \nabla f(x), d \rangle}{2\gamma_{\text{risk}} d^T \Sigma d}$$

In the Frank-Wolfe algorithm, iterates must remain within the feasible region (the unit simplex). Therefore, the step size γ is non-negative ($\gamma \geq 0$) and bounded above by the maximum allowed step size $\gamma_{\max} \leq 1$:

$$\gamma = \max\left(0, \min\left(\frac{-\langle \nabla f(x), d \rangle}{2\gamma_{\text{risk}} d^T \Sigma d}, \gamma_{\max}\right)\right)$$

4.2.2 Diminishing Step Size

The diminishing strategy [Jaggi, 2013] instead uses a predetermined sequence of step sizes, typically

$$\gamma = \frac{2}{k+2}.$$

Unlike ELS, this rule does not solve an additional minimisation problem at each iteration, having a step size relatively large during the first iterations that gradually decreases as k increases.

This makes the method simpler and computationally cheaper per iteration, but the step size does not directly adapt to the curvature of the objective function, where curvature refers to how rapidly the objective function changes as the portfolio weights move in a given direction. As a consequence, the FW gap may oscillate more strongly between consecutive iterations, especially in the early stages of the optimisation.

As the step size becomes smaller, the updates become more conservative and the iterates progressively stabilise near the optimum.

4.2.3 Stopping Criterion via Duality

By solving its linear subproblem at each iteration, the Frank-Wolfe algorithm directly produces a dual lower bound on the optimal value and, consequently, an upper bound on the primal suboptimality [Jaggi, 2013].

Let $f : \mathcal{C} \to \mathbb{R}$ be a convex differentiable function defined over a compact convex set $\mathcal{C} \subset \mathbb{R}^n$. By the first-order characterisation of convexity, for any $\mathbf{x}^* \in \mathcal{C}$:

$$f(\mathbf{x}^*) \geq f(\mathbf{x}_k) + \langle \nabla f(\mathbf{x}_k), \mathbf{x}^* - \mathbf{x}_k \rangle$$

Rearranging provides an upper bound on the primal suboptimality $f(\mathbf{x}_k) - f(\mathbf{x}^*)$:

$$f(\mathbf{x}_k) - f(\mathbf{x}^*) \leq \langle \nabla f(\mathbf{x}_k), \mathbf{x}_k - \mathbf{x}^* \rangle$$

Since the optimal point $\mathbf{x}^*$ lies within the feasible set $\mathcal{C}$, the right-hand side is upper-bounded by taking the maximum over all feasible points $\mathbf{x} \in \mathcal{C}$. This corresponds to the Linear Minimisation Oracle (LMO) solution $\hat{\mathbf{x}}_k = \arg\min_{\mathbf{x}\in\mathcal{C}} \langle \nabla f(\mathbf{x}_k), \mathbf{x} \rangle$:

$$\langle \nabla f(\mathbf{x}_k), \mathbf{x}_k - \mathbf{x}^* \rangle \leq \max_{\mathbf{x}\in\mathcal{C}} \langle \nabla f(\mathbf{x}_k), \mathbf{x}_k - \mathbf{x} \rangle = \langle \nabla f(\mathbf{x}_k), \mathbf{x}_k - \hat{\mathbf{x}}_k \rangle$$

We define this quantity as the duality gap $g(\mathbf{x}_k)$:

$$g(\mathbf{x}_k) = \langle \nabla f(\mathbf{x}_k), \mathbf{x}_k - \hat{\mathbf{x}}_k \rangle = \max_{\mathbf{x}\in\mathcal{C}} \langle \nabla f(\mathbf{x}_k), \mathbf{x}_k - \mathbf{x} \rangle$$

Hence, $g(\mathbf{x}_k) \geq f(\mathbf{x}_k) - f(\mathbf{x}^*)$. Because the duality gap bounds the primal suboptimality from above, enforcing $g(\mathbf{x}_k) \leq \epsilon$ guarantees that $f(\mathbf{x}_k) - f(\mathbf{x}^*) \leq \epsilon$, making $g(\mathbf{x}_k) \leq \epsilon$ a suitable stopping criterion.

Geometrically, a large duality gap indicates that the tangent hyperplane allows substantial room for cost reduction. Conversely, a small gap signifies that limited first-order improvement remains available. Ultimately, when $g(\mathbf{x}_k) = 0$, the iterate $\mathbf{x}_k$ satisfies the necessary and sufficient first-order optimality conditions, confirming global optimality for the constrained convex problem.

4.3 Away-Step Frank-Wolfe (ASFW)

Since the Standard FW algorithm has a sublinear convergence rate of $\mathcal{O}(1/k)$ it experiences a phemomenon known as zig-zagging when the optimal solution x^* lies on the boundary of $\mathcal{C}$. Standard FW directions are always directed towards extreme points; however, when near the optimum, these directions become increasingly orthogonal to the gradient, causing slow progress and they continuously oscillate between extreme points adding weight to the vertices that are no longer optimal.

To overcome this limitation, a new algorithm known as the Away-Step Frank-Wolfe (ASFW) algorithm was introduced. ASFW enables movement away from suboptimal active vertices. It does this by maintaining x_k as an explicit convex combination of active vertices and actively reducing their weights.

The first direction is the standard FW Direction (d_k^{FW}), which moves the current point towards a new candidate extreme point $\hat{x}_k^{FW} = \arg\min_{x\in\mathcal{C}} \nabla f(x_k)^\top (x - x_k)$ that minimises the linearised objective. The standard Frank-Wolfe direction is then defined as

$$d_k^{FW} = \hat{\mathbf{x}}_k^{FW} - x_k$$

This attempts to push the iterate toward a vertex that promises the steepest global reduction according to the first-order Taylor expansion at x_k.

The second direction computed at each iteration is the Away-Step direction d_k^{AS}. Unlike d_k^{FW}, which considers the entire feasible domain $\mathcal{C}$, d_k^{AS} focuses only on the active set of vertices $S_k \subset \text{vertices}(\mathcal{C})$ currently used to represent x_k as a convex combination ($x_k = \sum_{v\in S_k} \alpha_v^{(k)} v$). The vertex $\hat{\mathbf{x}}_k$ is selected from this active set by maximising the directional derivative: $\hat{\mathbf{x}}_k = \arg\max_{v\in S_k} \langle \nabla f(x_k), v - x_k \rangle$. The Away-Step direction is then defined as the direction from x_k away from the selected active vertex: $d_k^{AS} = x_k - \hat{\mathbf{x}}_k$

The away-step direction points away from the worst vertex (the one with the highest value of the linearised function) describing the current iterate. It does this by shifting mass off its corresponding weight $\alpha_{\hat{\mathbf{x}}_k}^{(k)}$, directly counteracting the zig-zagging effect in standard FW.

However, there exist a few disadvantages of this approach. There is a high cost associated with storing and updating S_k. Additionally, representing iterate k as a combination of vertices is not necessarily unique.

Algorithm 3 Away-Step Frank-Wolfe Method [Lacoste-Julien and Jaggi, 2015]

Choose an initial point $x_1 \in C$ and set S_1 as its active set.
for $k = 1, 2, \ldots$ **do**
  Compute the Frank-Wolfe vertex:

$$\hat{x}_k^{FW} \in \arg\min_{x \in C} \nabla f(x_k)^\top (x - x_k).$$

  Set $d_k^{FW} = \hat{x}_k^{FW} - x_k$.
  if $\nabla f(x_k)^\top d_k^{FW} = 0$ **then**
    STOP and return x_k.
  end if
  Compute the away vertex:

$$\hat{x}_k^{AS} = \arg\max_{x \in S_k} \nabla f(x_k)^\top (x - x_k).$$

  Set $d_k^{AS} = x_k - \hat{x}_k^{AS}$.
  if $\nabla f(x_k)^\top d_k^{FW} \leq \nabla f(x_k)^\top d_k^{AS}$ **then**
    Set $d_k = d_k^{FW}$ and $\bar{\alpha}_k = 1$.
  else
    Set $d_k = d_k^{AS}$ and

$$\bar{\alpha}_k = \max\left\{\beta \geq 0 : x_k + \beta d_k^{AS} \in C\right\}.$$

  end if
  Choose step size $\alpha_k \in (0, \bar{\alpha}_k]$ suitably, and update:

$$x_{k+1} = x_k + \alpha_k d_k.$$

  Update the active set S_{k+1} containing the currently used vertices.
end for

4.4 Pairwise Frank-Wolfe (PWFW)

Another variation of the Frank-Wolfe algorithm is the Pairwise Frank-Wolfe (PWFW) method. Rather than moving toward a single corner of the domain, PWFW performs a single update step that transfers weight directly from an away vertex to a newly selected Frank-Wolfe vertex.

At each iteration, the search direction is:

$$d_k = \hat{x}_k^{FW} - \hat{x}_k^{AS}.$$

This direction simultaneously adds mass to the Frank-Wolfe vertex and removes the away vertex. The maximum step size is constrained by the weight of the away vertex:

$$\bar{\alpha}_k = \alpha_{\hat{x}_k^{AS}}^{(k)},$$

which guarantees that $x_k + \alpha_k d_k \in C$ for any $\alpha_k \in [0, \bar{\alpha}_k]$.

Algorithm 4 Pairwise Frank-Wolfe Method [Lacoste-Julien and Jaggi, 2015]

Choose an initial point $x_1 \in C$ and set S_1 as its active set.
for $k = 1, 2, \ldots$ **do**
  Compute the Frank-Wolfe vertex:

$$\hat{x}_k^{FW} = \arg\min_{x \in C} \nabla f(x_k)^\top (x - x_k).$$

  Compute the away vertex:

$$\hat{x}_k^{AS} = \arg\max_{x \in S_k} \nabla f(x_k)^\top (x - x_k).$$

  Set $d_k = \hat{x}_k^{FW} - \hat{x}_k^{AS}$.
  if $\nabla f(x_k)^\top (\hat{x}_k^{FW} - x_k) = 0$ **then**
    STOP and return x_k.
  end if
  Set maximum step size $\bar{\alpha}_k = \alpha_{\hat{x}_k^{AS}}^{(k)}$.
  Choose step size $\alpha_k \in (0, \bar{\alpha}_k]$ suitably, and update:

$$x_{k+1} = x_k + \alpha_k d_k.$$

  Update the active set S_{k+1} containing the currently used vertices.
end for

4.5 Convergence Properties

The standard Frank-Wolfe method achieves a sublinear convergence rate of $\mathcal{O}(1/k)$ [Jaggi, 2013] as established by the Frank-Wolfe convergence theorem due to the use of a LMO rather than full projections on $\mathcal{C}$. When the optimal solution x^* lies on the boundary of the feasible set, the iterates begin to zig-zag between the vertices. The algorithm cannot remove poorly performing vertices from the active set, so it continues to oscillate, making only slow progress.

This sublinear bound is tight in the general convex setting; however, modifying the step search space alters these theoretical limits dramatically. Under σ-strong convexity over polyhedral sets, both Away-Step and Pairwise variants bypass this, establishing a linear convergence rate. This acceleration is achieved without sacrificing the low per-iteration memory and computational cost that makes first-order conditional gradient methods scalable.

For the unit simplex $\mathcal{C}$, the simplex LMO has cost $\mathcal{O}(n)$, active-set variants preserve sparsity while significantly reducing overall runtime. By permitting weight reduction on suboptimal vertices, ASFW and PWFW avoid dense vertex accumulation, offering a compelling balance between sparse support maintenance and rapid numerical progress.

5 Description of Datasets

To evaluate the numerical performance of the Frank-Wolfe variants, three benchmark portfolio selection datasets constructed from real-world weekly asset returns were analysed. These were Dow Jones, FTSE 100, and S&P 500 [Bruni et al., 2016].

The Dow Jones dataset contains 28 assets and 1363 weekly observations, the FTSE 100 dataset contains 83 assets and 717 weekly observations, while the $S\&P$ 500 dataset contains 442 assets and 595 weekly observations. This provides three increasingly high-dimensional portfolio optimisation problems, allowing the performance of the different Frank-Wolfe variants to be evaluated as the number of assets increases.

The Dow Jones dataset has the longest time series, with 1363 weekly observations, whereas the FTSE 100 and $S\&P$ 500 datasets contain 717 and 595 observations respectively. These are in years, 26, 14

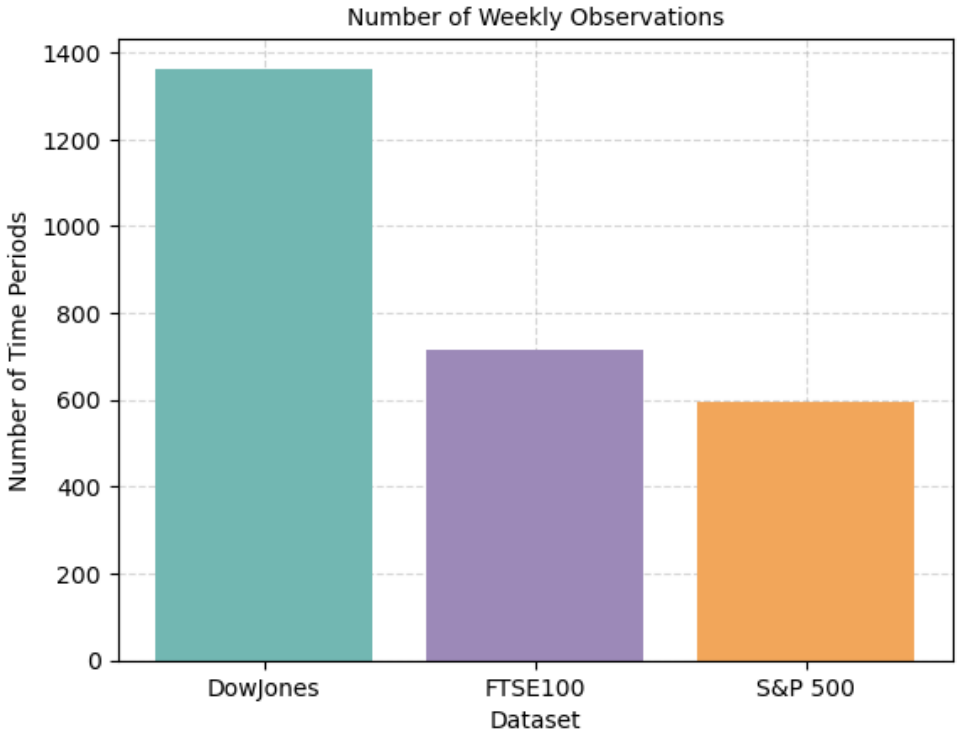


Figure 1: Number of Weekly Observations Per Dataset

and 11 respectively approximately. This difference is relevant when considering the estimation of the expected return vector and covariance matrix, since these quantities are estimated from the available historical observations.

5.1 Return Characteristics

An examination of the statistical properties of average asset returns across the three datasets reveals several important distributional characteristics that are relevant to the subsequent portfolio optimisation analysis. As summarised in Table 1, the mean return for each individual asset was first calculated across the available time periods. Descriptive statistics were then computed across these asset-level average returns, including the mean, median, standard deviation, minimum, maximum, skewness, and kurtosis.

Table 1: Descriptive Statistics of Average Asset Returns

Dataset	Assets	Mean	Median	Std.	Min.	Max.	Skew.	Kurt.
Dow Jones	28	0.002885	0.002506	0.001258	0.001278	0.006054	1.288624	0.671913
FTSE 100	83	0.002508	0.002507	0.001506	-0.000610	0.008018	0.807218	1.337827
S&P 500	442	0.002546	0.002339	0.001471	-0.000626	0.010325	1.499171	4.517561

Average asset returns are broadly similar across the three datasets, ranging from 0.2508% for the FTSE 100 to 0.2885% for the Dow Jones. The FTSE 100 has the highest cross-sectional standard deviation (0.001506), while the S&P 500 has the widest range of average returns, this is unsurprising as it has the most assets, and the strongest positive skewness (1.4992) and kurtosis (4.5176). The results indicate similar average returns but meaningful differences in dispersion, asymmetry, and tail behaviour, which may influence portfolio allocations.

5.2 Correlation

In addition to individual asset dynamics, the pairwise relationships between assets help to define the quadratic objective's underlying matrix structure.

Table 2: Correlation Summary Across Datasets

Dataset	Mean Correlation	Median Correlation	Minimum Correlation	Maximum Correlation
Dow Jones	0.336582	0.325762	0.100503	0.787045
FTSE 100	0.345084	0.338179	-0.089827	0.845918
S&P 500	0.412351	0.406816	-0.008174	0.984175

As shown in Table 2, the S&P 500 has the highest mean correlation (0.412), meaning its 442 assets tend to move together more than those in the Dow Jones or FTSE 100. This is particularly relevant to portfolio optimisation because the covariance matrix captures these relationships when the Frank-Wolfe algorithm determines the portfolio weights.

The minimum and maximum correlations are also informative. For example, the S&P 500 ranges from approximately -0.008 to 0.984, showing that some assets are almost uncorrelated, while others move extremely closely together. High correlations are likely to be due to these assets belonging to the same industry (such as oil companies) and so are subject to similar drivers.

6 Results

To assess the practical efficiency and convergence behaviour of the conditional gradient methods, the variants of FW were evaluated using the complete historical sample covariance matrix Σ and mean return vector $\bar{\mathbf{r}}$ derived from each benchmark dataset. Optimisation was performed using a risk-aversion parameter of $\gamma_{risk} = 1.0$. A γ_{risk} value of 0.01 was also tested and as expected the variance impact is marginal and the algorithm ends up with one asset, which is the asset with the highest expected return. These were asset 17 for Dow Jones, asset 77 for FTSE 100 and asset 86 for S&P 500. The Frank-Wolfe variants were executed with a duality gap convergence tolerance of 10^{-6}. We initially used a maximum iteration value of 1000; however, we settled with a value of 10000 as some algorithms exceeded 1000 iterations. Also, the algorithms need a starting portfolio. We have set this as allocating 100% to the first asset, asset 0.

Table 3: Comparative Performance of Optimisation Algorithms

Dataset	**Algorithm**	**n**	**T**	**Iterations**	**Final Objective**	**Final Gap**	**Active Size**	**Time (s)**
Dow Jones	FW	28	1363	29	-0.004183	9.447e-07	4	0.002036
Dow Jones	ASFW	28	1363	9	-0.004183	6.308e-08	4	0.001118
Dow Jones	PWFW	28	1363	13	-0.004183	3.594e-07	4	0.001000
FTSE 100	FW	83	717	4172	-0.005239	9.940e-07	5	0.365267
FTSE 100	ASFW	83	717	11	-0.005240	3.356e-07	4	0.002317
FTSE 100	PWFW	83	717	12	-0.005240	9.412e-07	4	0.001014
S&P 500	FW	442	595	3829	-0.007494	9.986e-07	6	0.554001
S&P 500	ASFW	442	595	16	-0.007494	7.699e-07	5	0.001068
S&P 500	PWFW	442	595	14	-0.007494	1.114e-07	5	0.002162

6.1 Convergence Rate

The convergence behaviour differs substantially across the algorithms as the dimension of the asset universe n increases, as illustrated in Table 3 and Figure 2.

For the smallest dataset, Dow Jones (28 assets), all the conditional gradient methods converge quickly within the specified duality gap tolerance. Standard FW takes 29 iterations, while ASFW and PWFW reach optimality in 9 and 13 iterations respectively.

As the problem dimensionality increases, standard FW exhibits a substantial increase in the number of iterations required for convergence. On both the FTSE 100 (83 assets) and S&P 500 (442 assets) datasets, FW exceeds the initial 1000-iteration budget without satisfying the specified duality gap tolerance. After increasing the maximum iteration limit to 10000, FW requires 4172 iterations on

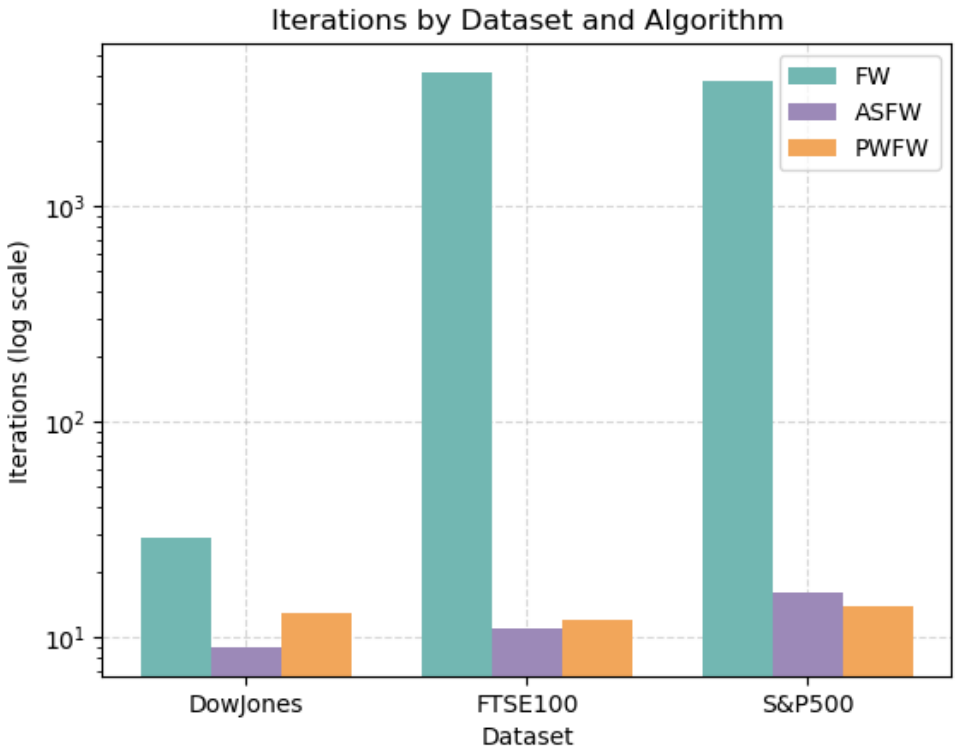


Figure 2: Iterations by Dataset and Algorithm

FTSE 100 and 3829 iterations on the S&P 500 dataset to reach the convergence criterion. This is due to the classic sublinear decay rate $\mathcal{O}(1/k)$ associated with the standard FW method, alongside severe boundary "zig-zagging" between active vertices. As the optimisation approaches the boundary of the feasible simplex, these increasingly small updates limit the rate at which the method can reduce the duality gap, resulting in substantially more iterations for the larger portfolio problems.

Conversely, ASFW and PWFW exhibit substantially faster convergence than standard FW, consistently terminating in fewer than 20 iterations across all three datasets. On FTSE 100, ASFW and PWFW require 11 and 12 iterations, respectively, compared with 4172 for FW. On S&P 500, they require only 16 and 14 iterations, compared with 3829 for FW. This performance advantage is due to active-set management, which allows ASFW and PWFW to remove or reduce the weight of unfavourable vertices and avoid the slowdown experienced by standard FW.

6.2 Objective Value

The objective values are very similar across algorithms for each dataset, indicating that all methods identify comparable solutions. For Dow Jones, all three methods achieve a final objective of -0.004183. On FTSE 100, ASFW and PWFW achieve -0.005240, while FW reaches -0.005239. On S&P 500, all three algorithms achieve a final objective value of -0.007494. The small differences in some of the final objective values, together with the final duality gaps below 10^{-6}, indicate that the methods converge to highly similar solutions.

The objective values improve substantially as the asset universe expands, with S&P 500 achieving the best (most negative) objective of -0.007494. This indicates that larger asset universes offer greater optimisation opportunities, as the algorithm can select from a broader pool of assets to construct a more favourable risk-return profile. However, the improvement from FTSE 100 to S&P 500 is proportionally smaller than from Dow Jones to FTSE 100, suggesting diminishing returns to adding more assets beyond a certain threshold.

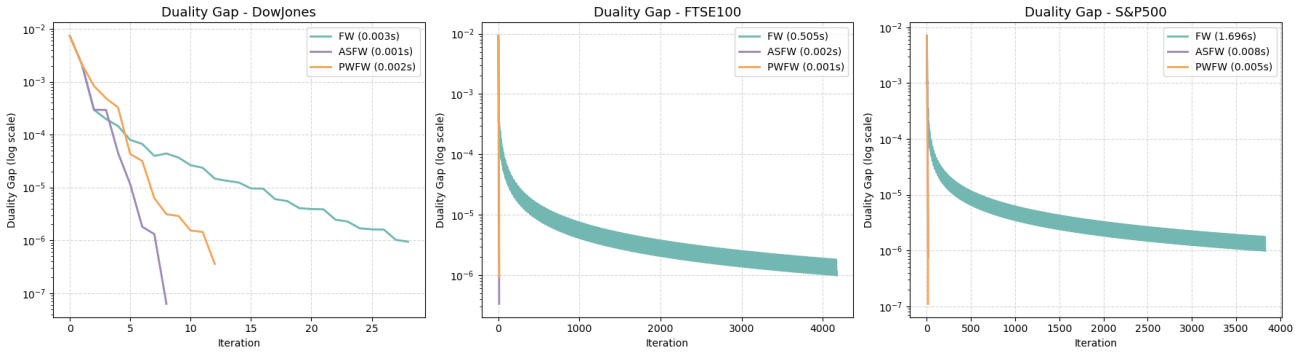


Figure 3: Duality Gap Per Iteration

6.3 Duality Gap

The duality gap provides additional insight into solution quality. ASFW achieves small final gaps of 6.308×10^{-8} for Dow Jones, 3.356×10^{-7} for FTSE 100, and 7.699×10^{-7} for S&P 500, all below the specified tolerance of 10^{-6}. PWFW also achieves gaps below the convergence tolerance, with final gaps of 3.594×10^{-7}, 9.412×10^{-7}, and 1.114×10^{-7} for Dow Jones, FTSE 100, and S&P 500, respectively. Standard FW reaches gaps of 9.447×10^{-7}, 9.940×10^{-7}, and 9.986×10^{-7} for the three datasets, respectively.

Figure 3 illustrates the convergence behaviour of the duality gap across iterations for the different algorithms and datasets.

The results show that ASFW and PWFW reduce the duality gap substantially faster than standard FW. Standard FW exhibits signs of "zig-zagging" finding it hard to converge as it requires substantially more iterations on the larger FTSE 100 and S&P 500 datasets. Meanwhile, both variants reach the 10^{-6} tolerance within fewer than 20 iterations across all datasets. This comparison reinforces the statement that the away-step and pairwise mechanisms enable significantly faster convergence by removing suboptimal vertices from the active set, allowing the algorithms to continue making progress where standard FW would be slow.

6.4 Final Active Set Size

The final active set sizes provide insight into the sparsity of the portfolios produced by the different FW variants. Across all datasets, the algorithms select only a small number of assets despite the substantially larger asset universes. For Dow Jones, all three methods identify the same four active assets, with very similar portfolio weights. FW, ASFW, and PWFW assign approximately 17.6%, 31.0%, 45.2%, and 6.1% to assets 0, 17, 18, and 21, respectively.

For FTSE 100, FW retains five active assets, whereas ASFW and PWFW reduce the active set to four assets. The additional asset selected by FW (asset 0) has a very small weight of only 0.057%, while the remaining four assets receive weights that are very similar across all three methods. This suggests that the away-step and pairwise variants produce a slightly sparser portfolio by eliminating an asset with negligible weight.

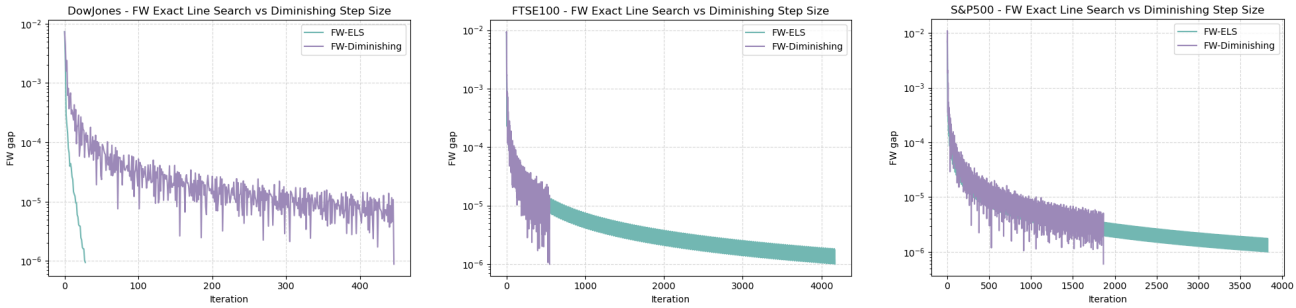
A similar pattern is observed for the S&P 500 dataset. FW produces a portfolio with six active assets, while ASFW and PWFW identify five. The additional asset retained by FW (asset 0) has a weight of approximately 0.60%, with the remaining five assets receiving very similar weights across the three methods. So, as above they eliminate an asset with negligible weight. The reason why asset 0 often appears in the active final set is likely due to our initial condition of setting 100% of our portfolio to asset 0.

6.5 Execution Time

The computational efficiency results show that the relative performance of the algorithms depends on the size of the optimisation problem. For the Dow Jones dataset, the differences in execution time are small, with ASFW and PWFW requiring approximately 0.001 seconds compared with 0.002 seconds for FW. For the larger FTSE 100 and S&P 500 datasets, however, the active-set methods provide a substantial reduction in computational time. On the FTSE 100, ASFW and PWFW require approximately 0.0023 and 0.0010 seconds respectively, compared with 0.365 seconds for FW. On the S&P 500, ASFW and PWFW require approximately 0.0011 and 0.0021 seconds respectively, compared with 0.554 seconds for FW.

These results suggest that the computational advantage of ASFW and PWFW becomes increasingly important as the asset universe grows. However, the results should not be interpreted as showing that the active-set variants are always faster, since their additional active-set management can introduce overhead on smaller problems.

6.6 Diminishing Step Size vs Exact Line Search



(a) Dow Jones (b) FTSE 100 (c) S&P 500

Figure 4: Frank-Wolfe convergence using exact line search and diminishing step size across the three datasets.

As previously discussed from a theoretical perspective, the experimental results confirm that the choice of step size strategy has a noticeable effect on the computational performance of the Frank-Wolfe algorithm.

On Dow Jones, ELS performs substantially better than the diminishing step size strategy, reaching the stopping tolerance in only 29 iterations compared to 447 iterations for the diminishing rule. Both methods identify 4 active assets and essentially the same final objective value of approximately -0.004183.

For FTSE 100, the opposite behaviour is observed. The diminishing step size reaches the stopping tolerance after 548 iterations, while ELS requires 4172 iterations. The difference is also evident in execution time with the diminishing strategy requiring approximately 0.010 seconds, compared with approximately 0.190 seconds for ELS.

A similar behaviour, as FTSE 100, is observed on S&P 500. The diminishing strategy converges in 1866 iterations, whereas ELS requires 3829 iterations. It is also faster in terms of execution time, requiring approximately 0.082 seconds compared with 0.351 seconds for ELS. Both methods achieve essentially the same final objective value of approximately -0.007494. However, the diminishing strategy obtains a smaller final duality gap of 6.005×10^{-7} compared with 9.986×10^{-7} for ELS. ELS identifies 6 active assets, while the diminishing strategy identifies 5.

All in all, the experimental results show that Exact Line Search is not uniformly superior to the diminishing step size strategy. Although ELS selects the step size that minimises the objective along the current Frank-Wolfe direction, this locally optimal choice does not necessarily result in faster global convergence. In our analysis, ELS performs particularly well on the smaller Dow Jones dataset, requiring substantially fewer iterations and less computational time. In contrast, the diminishing step

size strategy provides better practical performance on the larger FTSE 100 and S&P 500 datasets, requiring fewer iterations and less computational time while achieving comparable final objective values and, in the case of S&P 500, a smaller final duality gap.

Table 4: Comparison of Frank-Wolfe - ELS vs Diminishing Step Size

Dataset	Method	Active Assets	Time (s)	Iterations	Final Objective	Final Gap
Dow Jones	FW-ELS	4	0.000998	29	-0.004183	9.447×10^{-7}
	FW-Diminishing	4	0.007973	447	-0.004183	8.847×10^{-7}
FTSE 100	FW-ELS	5	0.190003	4172	-0.005239	9.940×10^{-7}
	FW-Diminishing	4	0.009968	548	-0.005240	9.940×10^{-7}
S&P 500	FW-ELS	6	0.351352	3829	-0.007494	9.986×10^{-7}
	FW-Diminishing	5	0.081686	1866	-0.007494	6.005×10^{-7}

6.7 Comparison with Projected Gradient method

We also conducted an experiment comparing the Projected Gradient Descent (PGD) method with the Frank-Wolfe variants to evaluate their computational efficiency and solution accuracy and sparsity.

A key difference between the two approaches is the cost of enforcing feasibility. Each PGD iteration requires a Euclidean projection onto the simplex, whereas FW requires only a Linear Minimisation Oracle (LMO). Efficient simplex projection methods are described by Condat [Condat, 2016]. Also as shown in Table 6, simplex projection is approximately 7.50, 8.40, and 8.49 times more expensive than the LMO for Dow Jones, FTSE 100, and S&P 500, respectively. Although the relative difference does not increase substantially between FTSE 100 and S&P 500, the projection remains consistently more computationally expensive than the LMO.

Despite the additional cost of projection at each iteration, PGD achieves accurate solutions across all three datasets. As shown in Table 5, the final duality gaps are 7.775×10^{-9} for Dow Jones, 8.854×10^{-8} for FTSE 100, and 5.054×10^{-7} for S&P 500. The final objective values are also consistent across the datasets, with values of -0.004183, -0.005240, and -0.007494, respectively.

In terms of convergence, PGD requires only 102 iterations for Dow Jones, 310 iterations for FTSE 100, and 1007 iterations for S&P 500. The corresponding execution times are approximately 0.005, 0.014, and 0.088 seconds. The increase in both iterations and computational time as the number of assets grows is particularly evident for the S&P 500 dataset.

Table 5: Projected Gradient Descent results for the three datasets.

Dataset	Num. Assets	Time Periods	Iterations	Final Objective	Final Duality Gap	Active Assets	Time (s)
Dow Jones	28	1363	102	-0.004183	7.775×10^{-9}	4	0.004983
FTSE 100	83	717	310	-0.005240	8.854×10^{-8}	4	0.014091
S&P 500	442	595	1007	-0.007494	5.054×10^{-7}	5	0.088026

The results also demonstrate that PGD can produce sparse solutions after applying the specified tolerance, with only 4 active assets for Dow Jones and FTSE 100 and 5 for the S&P 500. This is comparable with the sparse solutions produced by the FW variants.

In general, the results indicate that FW variants provide an alternative to PGD that is more computationally efficient for this portfolio optimisation problem. The main advantage of FW is that feasibility is maintained through the linear minimisation oracle rather than an explicit simplex projection. The computational advantage becomes particularly relevant for larger asset universes, although the total performance of an algorithm depends on both its per-iteration cost and the number of iterations required.

Table 6: Comparison between the Linear Minimisation Oracle and simplex projection : Ratios are computed using unrounded timings

Dataset	Num. Assets	LMO Time (s)	Projection Time (s)	Projection / LMO
Dow Jones	28	2.0×10^{-6}	1.3×10^{-5}	7.499
FTSE 100	83	2.0×10^{-6}	1.5×10^{-5}	8.398
S&P 500	442	2.0×10^{-6}	2.0×10^{-5}	8.489

6.8 Restricting to last 595 Time Periods

We note that the data sets are not of equal lengths. To ensure a fair comparison, all datasets were therefore limited to their most recent 595 observations. This would mean all 3 datasets are taken from the same data period and so are subject to the same market conditions.

The results in Table 7 reinforce the conclusions from the full historical samples. Standard FW remains substantially slower to converge as the number of assets increases, requiring 2698 iterations for FTSE 100 and 3829 for S&P 500, compared with fewer than 20 iterations for ASFW and PWFW. Despite this difference in convergence speed, all three methods achieve virtually identical objective values within each dataset.

The active sets also remain highly sparse relative to the size of the asset universes. Dow Jones produces two active assets for all methods, while FTSE 100 produces nine for FW and eight for ASFW and PWFW. For S&P 500, FW selects six assets, compared with five for ASFW and PWFW. Hence, the active-set variants generally produce equally or more sparse portfolios while requiring substantially fewer iterations.

Table 7: Portfolio optimisation results using the last 595 time periods.

Dataset	**Algorithm**	n	T	**Iterations**	**Objective**	**Duality Gap**	**Active Size**	**Time (s)**
Dow Jones	FW	28	595	2	-0.004419	-6.7763×10^{-20}	2	0.000000
Dow Jones	ASFW	28	595	2	-0.004419	-6.7763×10^{-20}	2	0.001177
Dow Jones	PWFW	28	595	2	-0.004419	-6.7763×10^{-20}	2	0.000000
FTSE 100	FW	83	595	2698	-0.004434	9.7880×10^{-7}	9	0.290244
FTSE 100	ASFW	83	595	20	-0.004434	7.4270×10^{-7}	8	0.002029
FTSE 100	PWFW	83	595	18	-0.004434	8.8182×10^{-7}	8	0.002028
S&P 500	FW	442	595	3829	-0.007494	9.9862×10^{-7}	6	0.500187
S&P 500	ASFW	442	595	16	-0.007494	7.6988×10^{-7}	5	0.001994
S&P 500	PWFW	442	595	14	-0.007494	1.1141×10^{-7}	5	0.001993

Note: Although the duality gap is theoretically non-negative, the implementation may return very small negative values due to floating-point numerical precision. In particular, the value -6.7763×10^{-20} for the Dow Jones dataset is numerically indistinguishable from zero and should therefore be interpreted as a zero duality gap.

7 Further Analysis

7.1 Rolling-Window Portfolio Optimisation

We also tested the three algorithms, Frank-Wolfe (FW), Away-Step Frank-Wolfe (ASFW), and Pairwise Frank-Wolfe (PWFW) using a rolling-window approach on the Dow Jones, FTSE 100, and S&P 500 datasets. To ensure a consistent comparison, all three datasets were restricted to their most recent 595 observations. This ensures that the datasets cover the same historical period and are therefore exposed to broadly comparable market conditions. At each rebalancing point, the previous 52 observations were used to estimate the mean return and covariance matrix, which were then used to construct a no-short-selling portfolio.

7.1.1 Risk Parameter Analysis

We tested six different values of the risk-aversion parameter, $\gamma = 0.01, 0.1, 0.5, 1, 5, 10$. The parameter γ controls the relative importance assigned to portfolio variance in the objective function. Smaller values of γ place relatively greater emphasis on expected return, whereas larger values place greater emphasis on reducing portfolio risk.

The optimised portfolio was held for the subsequent four periods, after which the rolling window was moved forward by four periods and the portfolio was re-optimised using the newly available observations. This process was repeated throughout the dataset, resulting in portfolio rebalancing every four periods.

To examine the resulting investment behaviour over time, the optimised portfolio weights were used to calculate the realised portfolio return over each four week holding period. Starting from an initial budget of 1, as specified in the Markowitz formulation, the portfolio value was updated according to

$$B_{t+1} = B_t(1 + r_{p,t}),$$

where B_t denotes the portfolio budget at period t and $r_{p,t}$ is the realised portfolio return obtained from the optimised weights. Repeating this process across the rolling windows produces a historical budget-evolution series. These series were plotted for different values of γ, allowing the effect of the risk-aversion parameter on the evolution of portfolio wealth to be examined for FW, ASFW, and PWFW.

For presentation purposes, Figure 5 shows the wealth trajectories obtained using PWFW only. The three algorithms produce very similar wealth trajectories for a given value of γ, so plotting all three methods resulted in substantial overlap and make the effect of the risk-aversion parameter more difficult to distinguish. PWFW is therefore used as a representative method for visualising the effect of γ.

This analysis assumes zero transaction costs when rebalancing the portfolio. In practice, buying and selling assets has costs, including the difference between buying and selling prices (the bid-ask spread), as well as potentially other transaction costs. Consequently, frequent rebalancing would reduce the realised portfolio wealth compared with the results presented here. The budget-evolution results therefore should be interpreted as a no-transaction-cost benchmark rather than a direct estimate of investable returns.

From Figure 5, we can see that the effect of the risk-aversion parameter γ varies across the three datasets. To select a representative value of γ for each dataset, we considered the value that produced the highest final portfolio wealth. It is important to note that these values are not necessarily the optimal risk levels but rather, they represent the best-performing γ values based on final out-of-sample portfolio wealth. The resulting values are shown in Table 8.

Table 8: Best-performing risk-aversion parameter and corresponding portfolio performance.

Dataset	**Best γ**	**Best Method**	**Final Wealth**
Dow Jones	0.01	FW	5.171
FTSE 100	0.01	FW	9.083
S&P 500	5.00	PWFW	6.922

The results show that the risk-aversion parameter associated with the highest realised wealth differs across the three datasets. For both the Dow Jones and FTSE 100, $\gamma = 0.01$ produces the highest final wealth. This corresponds to a relatively return-focused investor, placing comparatively little emphasis on portfolio variance. In contrast, the S&P 500 achieves its highest final wealth with $\gamma = 5$, representing a more risk-averse portfolio.

This difference shows that a lower risk aversion (lower γ) does not always produce higher returns, as might be expected. Rather, it reflects the interaction between the estimated return-risk characteristics of each asset universe and the subsequent realised returns during the training period. Hence, the risk-aversion parameter that performs best is dependent on both the dataset and the market conditions

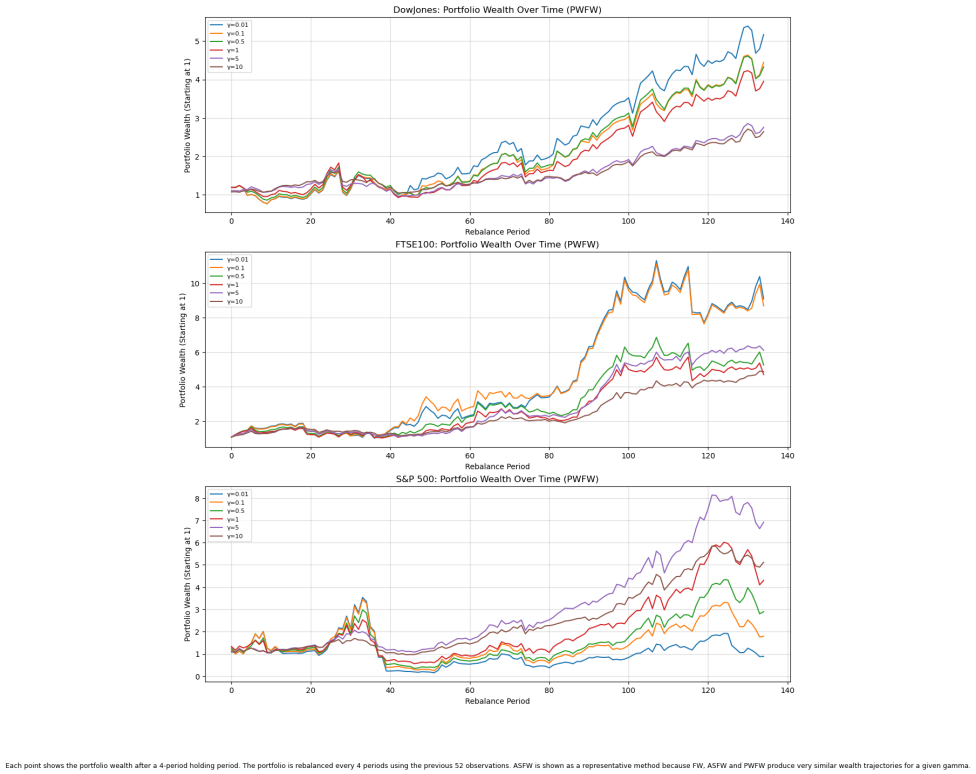


Figure 5: Portfolio Wealth Over Time

during the test period. Based on these results, $\gamma = 0.01$ is selected for the Dow Jones and FTSE 100, while $\gamma = 5$ is selected for the S&P 500 in the subsequent algorithm comparison.

7.1.2 Rolling-Window Optimisation Results

Table 9: Rolling-window optimisation results using the selected risk-aversion parameters.

Dataset	Algorithm	Mean Iter.	Mean Time (s)	Mean Obj.	Mean Gap	Mean Active
Dow Jones	FW	1.607	0.000057	-0.010047	-1.02×10^{-21}	1.007
Dow Jones	ASFW	1.607	0.000032	-0.010047	-1.02×10^{-21}	1.007
Dow Jones	PWFW	1.607	0.000076	-0.010047	-1.02×10^{-21}	1.007
FTSE 100	FW	2.067	0.000038	-0.016417	5.05×10^{-9}	1.022
FTSE 100	ASFW	2.037	0.000055	-0.016417	2.50×10^{-9}	1.022
FTSE 100	PWFW	2.030	0.000087	-0.016417	2.76×10^{-10}	1.022
S&P 500	FW	8910.452	0.696776	-0.011766	2.61×10^{-6}	9.156
S&P 500	ASFW	47.200	0.004179	-0.011767	7.52×10^{-7}	7.956
S&P 500	PWFW	35.059	0.002841	-0.011767	7.28×10^{-7}	7.963

Note: Again, although the duality gap is theoretically non-negative, the implementation may return very small negative values due to floating-point numerical precision. In particular, the value -1.02×10^{-21} for the Dow Jones dataset is numerically indistinguishable from zero and should therefore be interpreted as a zero duality gap.

The optimisation results show little difference between the algorithms for Dow Jones and FTSE 100, where all three methods converge in approximately two iterations per window. The differences become substantial for the S&P 500: FW requires over 8900 iterations on average, compared with 47 for ASFW and 35 for PWFW. PWFW is also the fastest of the three methods. The results therefore again show that the computational advantages of the active-set variants become particularly important as the dimensionality of the portfolio increases.

8 Conclusion

This project investigated the use of Frank-Wolfe optimisation methods for the classical Markowitz portfolio optimisation problem, with particular emphasis on the standard Frank-Wolfe (FW), Away-Step Frank-Wolfe (ASFW), and Pairwise Frank-Wolfe (PWFW) algorithms. The theoretical analysis established the convex quadratic formulation of the portfolio problem and demonstrated how the structure of the unit simplex allows the Frank-Wolfe linear minimisation oracle to be solved efficiently. The analysis also highlighted the limitations of standard FW, particularly its $O(1/k)$ convergence rate and the zig-zagging behaviour that can occur near boundary solutions.

The numerical results showed that ASFW and PWFW can significantly improve convergence, particularly for larger datasets, while producing objective values and sparse portfolios comparable to standard FW. While these sparse solutions demonstrate strong algorithmic efficiency, an active set of only around four assets may present a practical limitation for investors who prioritise broader portfolio diversification to mitigate non-systematic risk.

The step size analysis showed that exact line search is not consistently superior to diminishing step sizes, with performance depending on the dataset. The comparison with Projected Gradient Descent (PGD) further demonstrated the computational advantage of FW methods by avoiding the more expensive simplex projection.

Finally, the rolling-window analysis used the most recent 595 observations for all datasets to ensure a common historical period. The best-performing risk-aversion parameter was $\gamma = 0.01$ for Dow Jones and FTSE 100, and $\gamma = 5$ for the S&P 500, showing that the preferred risk level varies across datasets. The algorithms performed similarly for Dow Jones and FTSE 100, while for the S&P 500, FW required over 8900 iterations compared with 47 for ASFW and 35 for PWFW. This highlights the increasing computational advantage of the active-set variants for higher-dimensional problems.

9 LLM Usage Disclosure Statement

Generative LLM tools were restricted to support for individual study, Python library documentation lookups, and visual plotting/formatting.

References

[Bruni et al., 2016] Bruni, R., Cesarone, F., Scozzari, A., and Tardella, F. (2016). Real-world datasets for portfolio selection and solutions of some stochastic dominance portfolio models. *Data in Brief*, 8:858–862.

[Condat, 2016] Condat, L. (2016). Fast projection onto the simplex and the ℓ_1 ball. *Mathematical Programming*, 158(1–2):575–585.

[Jaggi, 2013] Jaggi, M. (2013). Revisiting Frank–Wolfe: Projection-free sparse convex optimization. In *Proceedings of the 30th International Conference on Machine Learning (ICML)*, volume 28, pages 427–435.

[Lacoste-Julien and Jaggi, 2015] Lacoste-Julien, S. and Jaggi, M. (2015). On the global linear convergence of Frank–Wolfe optimization variants. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 28, pages 496–504.